

Glass ceiling unshattered: asymmetric gender culture among immigrants in different German tracks

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Abstract

This paper traces the math gender gap among immigrant adolescents in the German three-track school system between ages 13 and 16. Using national education panel studies that record evolution of innate skills, I extract three stylized facts. The math gender gap widens in every track. In Gymnasium it widens more for immigrant girls than for native girls. In comprehensive schools the pattern reverses; immigrant girls are roughly indistinguishable from immigrant boys. I push the result with a matched sample on pre-tracking scores and with endogenous structural break of Penn's piecewise threshold across eight demographic cells. Gymnasium immigrant girls are the one cell that exhibits Penn's aspiration trap. And this is due to leniency from teachers towards immigrant girls. As such, immigrant girls allocate over-investment toward social competence rather than academic effort whereas boys' over-investment is allocated in academic excellence as teachers are biased against them.

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1 Introduction

Immigrant families carry an aspiration premium. They consistently hold higher educational ambitions for their children than comparable native families, after controlling for socioeconomic background (Kao and Tienda, 1995; Feliciano and Lanuza, 2017; Salikutluk, 2016). The premium is real and substantial. Immigrant boys who reach Gymnasium in Germany outperform native boys at +0.22 standard deviations on math. But it is not gender-neutral. Does the same premium reach the daughter, and if not, where does it go?

The answer depends on the institutional environment. The German three-track system sorts students at the end of primary school into the academic Gymnasium, the vocational Realschule, or the mixed Gesamtschule. Teacher recommendations mediate the sort along socioeconomic background, gender, and immigration status (Dustmann et al., 2016; Jürges and Schneider, 2011). Once sorted, students face radically different peer environments, teacher expectations, and curricular demands. The aspiration literature predicts divergent effort dynamics: students whose aspirations are attainable invest effort, those facing an unbridgeable gap withdraw (Ray, 2006; Genicot and Ray, 2017; Penn, 2017). I argue that immigrant families' traditional gender norms produce a gendered aspiration threshold, a glass ceiling that activates only when the institutional environment demands sustained academic commitment.

I test this by pooling two NEPS cohorts and estimating value-added models of Grade 9 math conditional on Grade 7. The value-added specification is a difference-in-differences of the gender gap. Conditioning on Grade 7 achievement, the female coefficient captures the gap that develops during secondary school, net of everything that came before.

Three stylized facts stand out. First, the math gender gap widens in all three tracks; girls who scored identically to boys at G7 fall behind by G9, and demographic controls do not move it. Second, the gap is asymmetric by immigration status in opposite directions across tracks. In Gymnasium, immigrant girls fall behind faster than native girls. In comprehensive schools, immigrant girls are protected; the within-immigrant gender gap is effectively zero.

Third, socioeconomic status does not moderate the gap once immigration is controlled. The mechanism is cultural, not material.

Two further pieces of evidence sharpen the result. A matched sample balanced on pre-tracking G4 scores roughly doubles the Gymnasium gender penalty and strengthens the immigrant-girl interaction. Endogenous structural-break estimation of Penn’s (2017) piecewise threshold across eight demographic cells finds that Gymnasium immigrant girls are the only cell exhibiting the aspiration trap. Below the threshold scores lock in; above it they regress. In every other cell effort compounds above the threshold as the model predicts. The glass ceiling is a regime switch at the intersection of gender, immigration, and institutional context, not a uniform drag.

The mechanism behind the trap, I argue, runs through teacher behavior. Conditional on the test score, teachers grade immigrant students worse than natives in both math and German, but they grade girls better than boys, and the pro-girl premium is sharpest in German. For immigrant girls the two biases roughly cancel: a Note penalty for being immigrant offset by a Note premium for being female. For immigrant boys there is no offset. Boys face a clean academic penalty, and the rational response is to push harder on the academic dimension to overcome it. Girls face academic neutrality combined with a social-compliance reward, and the rational response is to push the social dimension where the institution is already paying them. Penn’s match-on-one-dimension logic does the rest. Immigrant boys allocate their over-investment to academic excellence; immigrant girls allocate theirs to social competence. The academic shortfall the paper documents is not a punishment imposed on the daughter. It is the equilibrium response to a teacher signal that rewards her where her brother is told to push.

The paper proceeds as follows. Section 2 reviews the aspiration, identity, immigrant-paradox, and tracking literatures. Section 3 develops the glass ceiling hypothesis. Section 4 lays out the theoretical framework. Sections 5 and 6 describe the data and empirical strategy. Section 7 presents the main value-added results, including the occupational-aspiration

evidence. Section 8 reports the matched-sample evidence. Section 9 reports the endogenous structural-break estimation of Penn’s threshold across eight demographic cells. Section 10 reports the teacher-bias regressions at G7. Section 11 discusses. Section 12 concludes.

2 Literature

This paper sits at the intersection of four literatures: the economics of aspirations and effort, identity and education, the immigrant aspiration paradox, and the institutional effects of educational tracking. I review these bodies of work as an integrated whole, building toward the theoretical gap that the glass ceiling hypothesis fills.

The economics of aspirations begins with the observation that effort is not monotonically increasing in ambition. Ray (2006) formalizes the key insight that aspirations are reference points, and the gap between current status and the aspiration threshold determines investment behavior. When the gap is moderate, individuals exert effort to close it; when the gap is too large, effort collapses because the threshold appears unreachable. Genicot and Ray (2017) embed this in a general equilibrium model where aspirations are socially determined, showing that the interaction between individual reference points and aggregate inequality produces self-reinforcing polarization. Dalton et al. (2016) extend the logic to poverty traps, demonstrating that behavioral biases in aspiration-setting compound material deprivation, so that the poor underinvest not because they are irrational but because their aspirations are endogenously compressed. Empirically, Orkin et al. (2023) demonstrate that a simple aspiration-setting workshop for poor Kenyan households raises investment and living standards, confirming that the aspiration channel is real and manipulable. Beaman et al. (2012) show that exposure to female leaders in Indian villages substantially closes the gender aspiration gap for both parents and adolescents, establishing that role models can shift the threshold.

A critical refinement comes from Penn (2017), who introduces a class-specific aspiration

threshold that depends on the social context. In her model, individuals who fall below the class-appropriate threshold compensate for low self-esteem by distorting value identifications and withdrawing effort from their innate comparative advantages. The result is a lock-in equilibrium in which students below the threshold do not gradually improve but instead freeze. This model generates a sharp empirical prediction that distinguishes it from the smooth aspiration-gap framework of Genicot and Ray (2017): there should be a piecewise discontinuity in the persistence of achievement, with high persistence below the threshold and low persistence above it. Section 7 tests this prediction directly.

The question of why students withdraw effort connects directly to the economics of identity. Akerlof and Kranton (2002) argue that identity, meaning the social categories to which individuals belong and the behavioral prescriptions attached to those categories, is a primary determinant of effort in school. Students invest in academics when doing so is consistent with their identity; they withdraw when academic effort conflicts with group norms. Shayo (2009) generalizes this in a model where individuals choose group identities to maximize a combination of material payoffs and social status. The tracked school system makes this choice particularly stark because it concentrates peers who share the same institutional trajectory, forcing students to commit to an academic identity or retreat into an alternative peer group. The “acting white” literature operationalizes this tradeoff for minority students. Austen-Smith and Fryer Jr (2005) model a signaling equilibrium in which high-ability minority students face peer rejection for investing in education, because academic success signals departure from the in-group. Roland G. Fryer and Torelli (2010), using Add Health data, find that the phenomenon is real but contextual, strongest in integrated schools where the social cost of boundary-crossing is highest. Bisin et al. (2011) show that oppositional identities can persist in dynamic equilibrium when cultural segmentation is sufficiently high, creating a mechanism for intergenerational transmission of underinvestment.

The immigrant version of this identity conflict is distinct. Immigrant children do not face peer rejection for “acting white” in the same sense. Rather, they face family pressure when

academic commitment conflicts with gender-specific cultural expectations. Dahl et al. (2022) document this mechanism directly: granting birthright citizenship to immigrant children in Germany improves labor market integration for boys but backfires for girls, who are caught between host-country norms and parental cultural expectations. The mechanism is not peer-driven but family-driven, and it is gendered. Phalet and Schönpflug (2001) find that second-generation daughters in Turkish-German families shift toward egalitarian values, but sons remain as conservative as their fathers. The implication is that the family’s enforcement of traditional roles falls disproportionately on girls whose behavior deviates from expectations.

These identity dynamics interact with a robust empirical regularity: immigrant families hold higher educational aspirations for their children than comparable native families, even after controlling for socioeconomic background. Kao and Tienda (1995) established this “immigrant optimism” with U.S. data, showing that first- and second-generation youth outperform expectations based on their parents’ human capital. Feliciano and Lanuza (2017) demonstrate that the paradox persists intergenerationally, with children of immigrants maintaining higher aspirations than third-plus generation peers. In Germany specifically, Salikutluk (2016) shows that Turkish-origin students aim higher than natives at every level of prior achievement, driven by intergenerational transmission of upward mobility goals. Pomianowicz (2024) confirms this cross-nationally and identifies educational tracking as a key moderator: the aspiration-achievement gap is largest in stratified systems where high aspirations do not automatically translate into high-track placement. The paradox has a dark side, however. Tjaden and Hunkler (2017) argue that immigrant optimism can become an “optimism trap” in tracked systems, where families choose demanding tracks at lower GPAs, increasing non-completion risk. Becker et al. (2023), using NEPS data, track how immigrant aspirations evolve during lower secondary school and find that aspirations remain high but increasingly diverge from realistic expectations. Dollmann (2021) shows that positive ethnic choice effects exist across European countries but are strongest in systems with early, binding track decisions, precisely the institutional context that amplifies the consequences

of misaligned aspirations.

The cultural channel behind the math gender gap is now well established. Guiso et al. (2008), using PISA data across 40 countries, show that the gender gap in math disappears in more gender-equal cultures. Nollenberger et al. (2016) use the epidemiological approach with second-generation immigrants to isolate the cultural channel, finding that parental gender beliefs from the country of origin account for at least two-thirds of the math gender gap in the host country. Stoet and Geary (2018) document a “gender-equality paradox” in which STEM gender gaps are wider in more gender-equal countries, suggesting that increased choice amplifies pre-existing preference differences when material constraints are relaxed. On the supply side, Carlana (2019) shows that implicit teacher stereotypes, measured by Implicit Association Tests, widen the math gender gap by steering girls away from STEM-congruent tracks. In Germany, Tobisch and Dresel (2017) find that teachers assign lower expectations to students with Turkish-sounding names, and Jürges and Schneider (2011) show that boys and younger students receive systematically worse Gymnasium recommendations. Carlana et al. (2022) find that a mentoring program for high-achieving immigrant students improves aspirations and reduces segregation for boys but has limited effects on girls, a pattern consistent with the gendered ceiling this paper documents.

These cultural and identity dynamics play out within an institutional structure that amplifies inequality. The German three-track system is among the most stratified in the OECD. Hanushek and Woessmann (2006) establish that early tracking increases educational inequality without apparent aggregate performance gains. Brunello and Checchi (2007) confirm that earlier tracking strengthens the link between parental SES and student performance. Van de Werfhorst and Mijs (2010) synthesize the comparative evidence, concluding that between-school streaming, as practiced in Germany, is more segregating than within-school course-by-course tracking, because it bundles peer environment, teacher quality, and curricular demand into a single institutional package. For immigrants, tracking compounds disadvantage. Ruhose and Schwerdt (2016) show that tracking produces meaningfully neg-

ative effects on immigrant test scores relative to natives across reading, math, and science. Lüdemann and Schwerdt (2013) find that second-generation immigrants receive worse grades and track recommendations than comparable natives, though socioeconomic background accounts for much of the gap. Crul and Schneider (2010) argue that institutional arrangements, not individual characteristics, are the primary determinant of second-generation outcomes, with tracked systems being particularly consequential.

Yet the long-term effects are more ambiguous. Dustmann et al. (2016), exploiting cross-state variation in the age of tracking, find zero average long-term wage effects, suggesting that track misallocation washes out by adulthood. This null result, however, pools across demographic subgroups. The present paper argues that the aggregate null masks stark heterogeneity: tracking may have no average effect precisely because it helps some groups, such as immigrant boys in Gymnasium who benefit from the aspiration premium, while hurting others, such as immigrant girls who hit the glass ceiling. The contribution of this paper is to decompose the tracking effect along the gender-by-immigration margin and identify the aspiration mechanism that drives the divergence.

The four literatures supply the ingredients but do not combine them. The aspiration literature models effort as a function of the gap between current status and a threshold, but does not consider gendered thresholds. The identity literature models group-specific behavioral prescriptions, but treats immigrant identity as monolithic rather than gender-differentiated. The immigrant aspiration paradox documents high ambitions, but does not explain when the daughter's share of those ambitions is redirected away from the academic dimension. The tracking literature shows that institutional sorting amplifies inequality, but does not identify the channel through which it does so for immigrant girls. This paper combines all four. The glass ceiling is a gendered aspiration threshold carried by immigrant families that activates only when the institutional environment demands a level of academic commitment in tension with traditional gender roles. The mechanism is not discrimination, not material deprivation, and not peer rejection. It is the interaction between cultural gender

norms and institutional pressure.

3 Hypothesis: the glass ceiling in aspiring cultures

The literature above supplies two observations that, taken together, generate a sharp prediction. The first is the immigrant aspiration paradox. Immigrant families report higher educational aspirations for their children than comparable native families, after controlling for socioeconomic background (Feliciano and Lanuza, 2017). Their children translate this into higher ambitions. In Germany specifically, immigrant students aim higher than their non-immigrant counterparts despite facing systematic disadvantages at school (Pomianowicz, 2024). The positive immigrant coefficient in Gymnasium of +0.22 standard deviations is the direct empirical manifestation: immigrant boys in the academic track outperform native boys, on the back of the aspiration premium their families provide.

The second observation is that immigrant families carry traditional gender norms from their countries of origin into the host society, and these norms persist across generations. Parents from achievement-oriented cultures with traditional gender beliefs rate sons' math ability higher than daughters', and these parental beliefs predict children's self-concept (Nollenberger et al., 2016). Adolescents from immigrant groups in European countries hold more traditional gender beliefs than their majority peers. The norms do not suppress girls' aspirations uniformly. They impose a ceiling that activates when the institutional environment demands a level of academic identity commitment that conflicts with the family's gender expectations.

The glass ceiling metaphor, borrowed from the labor market literature, captures this. A glass ceiling is not a barrier at the bottom. It is an invisible constraint that binds only at the top. Immigrant girls in comprehensive schools, where academic demands are moderate and the peer environment is heterogeneous, do not hit the ceiling. The family's aspiration premium supports both sons and daughters because the level of academic commitment required

does not trigger a conflict with traditional gender roles. But in Gymnasium, where sustained academic investment is the price of admission to the peer group, the ceiling activates. The family keeps investing in the son's academic trajectory, which is culturally congruent with the male provider role, while redirecting the daughter's investment toward community obligations and domestic competence, or at minimum withdrawing the active encouragement that the son receives.

This is not a story about immigrant families being unsupportive. It is a story about support being conditional on the institutional demand. When the demand is moderate, immigrant culture supports both genders equally. When the demand is high, the daughter's investment is redirected away from academic effort toward the gender-appropriate alternative, and the academic outcome falls behind. The glass ceiling is unshattered not because immigrant families lack ambition, but because their ambition is gendered in ways that bind only in high-pressure academic environments.

Non-immigrant families do not exhibit this asymmetry. The gender gap among non-immigrants is roughly uniform across tracks because native German families, while subject to their own gender stereotypes, do not carry the same intensity of traditional gender-role enforcement. The aspiration premium is smaller, and the ceiling, if it exists, does not differentially activate by track.

The account generates three falsifiable predictions beyond the patterns the data already document.

1. Immigrants aspire more than natives in general.
2. Gymnasium immigrant girls aspire less than immigrant girls in the comprehensive track.
3. Parental educational aspirations for daughters relative to sons are lower among immigrant families in Gymnasium than in comprehensive schools.

4 Theoretical framework: gendered comparative advantage and Penn’s match-on-one-dimension

Penn (2017) models self-esteem as a function of the individual’s distance from a peer-defined “strong type” along a multidimensional skill vector. Skills are not interchangeable. The individual carries innate comparative advantages across the dimensions, and the strong type holds a high level on multiple dimensions at once. Self-esteem is preserved if the individual matches the strong type on at least one dimension, and corroded otherwise. The mechanism is cognitive dissonance. When the individual cannot match the strong type on every dimension, she picks the dimension where a match is feasible, over-invests there, and distorts the value of the chosen dimension upward to declare that the matched dimension is the one that matters. The dimensions on which she falls short are devalued. Effort is withdrawn from those dimensions even when one of them is her innate comparative advantage. The persistence function turns piecewise. Along the chosen dimension, performance above the strong-type threshold compounds; below it, no match is feasible on any dimension, and performance freezes.

The choice of which dimension to over-invest in is therefore the central object. In a peer environment that values multiple skills, the individual picks the dimension where she can plausibly clear the strong-type bar, and equilibrium investment is concentrated there. Two features of the choice matter here. First, the choice is endogenous to her beliefs about where the comparative advantage lies. Second, those beliefs are shaped by the family and the peer environment, which signal what she is good at through differential encouragement. Penn’s piecewise dynamic is not just an individual cognitive process. It is a cognitive process scaffolded by the social environment that defines the relevant skill set.

For German adolescents in the tracking system, the relevant skill vector has at least two dimensions. One is academic competence, observed through math and German performance. The other is social competence, observed through compliance, attentiveness, agreeableness,

and family-aligned respectability. The strong type on the academic dimension is set by Gymnasium peers and curricular standards. The strong type on the social dimension is set by the family, the community, and the classroom-behavior expectations that translate social competence into legible classroom presentation.

Boys' comparative-advantage signal points to academic excellence. Modern German cultural norms do not impose sustained social-competence demands on adolescent boys. The family signals academic effort as the relevant skill, the teacher signals it, the peers signal it. Boys over-invest in the academic dimension and distort their value of academics upward. Penn's piecewise dynamic plays out in the academic dimension as the model intended. Boys above the academic threshold compound, boys below lock in.

Girls' comparative-advantage signal points to social competence. The family signals compliance and respectability as the gender-appropriate skill. The teacher rewards attentiveness, lack of disruption, and neat presentation, all of which sit on the social dimension. Peers within gendered friendship structures reinforce the same. Girls over-invest in the social dimension. The pattern that Section 10 documents, math IRT for girls declining through secondary school while the German Note accumulates a pro-girl premium, is the visible signature of that choice. Native girls' social over-investment is moderated by a weaker gender norm. Immigrant girls' is amplified by stronger family and community signals about what a daughter should be good at.

The empirical prediction is single-crossing in the academic dimension. The girl's academic outcome depends on whether the residual academic investment she gives, after over-investing in social competence, clears the academic threshold she faces. Boys in any track and native girls in any track over-invest in academics, so they approach the threshold from above. Immigrant girls in comprehensive over-invest in social competence, but the comprehensive academic threshold is low enough that the residual academic investment still clears it. Immigrant girls in Gymnasium over-invest in social competence and fall below a high academic threshold that the residual cannot clear. The signature in the data is the one Section 9 finds:

the only cell that exhibits Penn’s piecewise lock-in is Gymnasium-immigrant-female.

The teacher-bias regressions in Section 10 corroborate the framework rather than complicate it. Teachers grade an academic component derived from test-relevant performance and a behavioral-compliance component derived from classroom presentation. The pro-girl premium of -0.342 on the Gymnasium German Note is the teacher’s reward for social-competence over-investment. It is not discrimination against boys. It is recognition of the dimension on which girls have been investing. The premium is not the source of the academic shortfall. Girls do not under-perform academically because the teacher rewards them in the social dimension. They under-perform academically because their over-investment was allocated there to begin with. The premium is consistent with the choice rather than the cause of any cost. What the premium does is reinforce the comparative-advantage signal: the teacher confirms that social competence is institutionally valuable, the family’s signal is corroborated, and the choice is sustained.

The mechanism is not a punishment story. Three actors coordinate the girl’s comparative-advantage signal toward the social dimension. The family signals social competence as the gender-appropriate skill. The teacher rewards social-competence behavior with Note leniency. The peer environment in Gymnasium maintains a high academic threshold that requires sustained academic over-investment to clear. The girl, applying Penn’s match-on-one-dimension logic, sends her over-investment to social competence. The academic dimension does not get the over-investment boys give it, and the academic shortfall follows.

5 Data

5.1 The NEPS

The National Educational Panel Study (NEPS) is a large-scale longitudinal study of education and competence development in Germany. I use two starting cohorts:

- Starting Cohort 2, SC2: Students first assessed in Grade 4 in 2008, tracked into sec-

ondary school at Grade 5, and followed through Grades 7 and 9. Version 11.0.0. $N = 1,311$ in the analysis sample.

- Starting Cohort 3, SC3: Students first assessed in Grade 5 in 2010, already in secondary school, and followed through Grades 7 and 9. Version 13.1.0. $N = 2,193$ in the analysis sample.

Both cohorts administer standardized math competence tests at Grades 7 and 9 using the same IRT framework with common item anchoring, enabling pooled estimation.

5.2 Outcome variable

The dependent variable is the corrected Weighted Likelihood Estimate (WLE) of mathematical competence at Grade 9 (`mag9_sc1`). This is an Item Response Theory (IRT) score: a maximum-likelihood estimate of latent math ability, corrected for measurement error, centered near zero with standard deviation approximately one. The “9” refers to Grade 9. The lagged dependent variable (`mag7_sc1`) is the same measure at Grade 7. Because both use common item anchoring, scores are comparable across waves and cohorts.

5.3 Regressors

5.4 Sample

The analysis is restricted to West Germany, as the data only allows for this binary regional indicator and the West takes up most of the sample. I retain the three main tracks: Gymnasium, Realschule, and Comprehensive. Hauptschule and Förderschule are excluded for small N . Students must have valid G7 and G9 math scores and non-missing SES. The pooled sample contains $N = 3,504$ students.

Table 1: Variable Definitions

Variable	Definition
Math G7	IRT-scaled math WLE at Grade 7. Controls for prior achievement, encompassing innate ability, family background, primary school quality, and track selection.
Female	= 1 if female. The coefficient of interest: captures the gender gap in G9 conditional on G7 score.
Immigrant	= 1 if migration background. SC2: generation status > 0, variable p400500_g1. SC3: migration background indicator > 1, variable p400000_g1.
High SES	= 1 if household has 101+ books, categories 4–6 on the 6-point ordinal p34005a. Standard proxy for parental education and cultural capital in NEPS.
Female $\times$ Immigrant	Does the gender gap differ for immigrants?
Female $\times$ High SES	Does household SES buffer girls against falling behind?

Table 2: Descriptive Statistics by Track, Pooled SC2 + SC3

	<i>N</i>	% Female	% Immigrant	% High SES	Mean G7	Mean G9	SD G9
Comprehensive	424	54.0	22.9	62.0	−0.20	−0.56	1.11
Realschule	780	45.1	17.3	56.9	0.33	−0.36	0.95
Gymnasium	2,300	52.0	20.3	79.5	1.09	0.59	1.11
Total	3,504						

6 Empirical Strategy

6.1 Value-added specification

The baseline specification, estimated separately within each track j , is:

$$\text{mag9}_{ij} = \alpha_j + \beta_{1j} \cdot \text{mag7}_{ij} + \beta_{2j} \cdot \text{Female}_i + \gamma_j \cdot X_i + \epsilon_{ij} \quad (1)$$

where mag9_{ij} is the math WLE at Grade 9 for student i in track j , mag7_{ij} is the same at Grade 7, and X_i includes immigration status and SES.

6.2 Identification: the diff-in-diff logic

The specification is a difference-in-differences of the gender gap. Consider a boy and a girl who score identically at G7. If $\beta_2 < 0$, the girl scores lower at G9 despite starting at the same level. The first difference is across gender (boy vs. girl); the second difference is across time (G7 vs. G9), absorbed by conditioning on G7. The coefficient β_2 estimates the gender gap that develops during G7–G9, net of prior achievement. A negative β_2 means girls fall behind; a positive β_2 means girls catch up.

6.3 Progressive specifications

I estimate four models within each track:

1. Baseline: $G9 \sim G7 + \text{Female}$.
2. Demographics: $+ \text{Immigrant} + \text{High SES}$.
3. Immigration interaction: $+ \text{Female} \times \text{Immigrant}$.
4. Full: $+ \text{Female} \times \text{High SES}$.

The stability of the Female coefficient from model 1 to model 2 tests whether the gender gap is an artifact of compositional differences across demographic groups. The interactions in models 3 and 4 test whether the gap is moderated by immigration and SES, the channels that the aspiration literature identifies.

7 Results

Figure 1 provides the descriptive overview. The gender gap, measured as Cohen's d with male minus female, is plotted at G7 and G9 by track and immigration status. The gap widens from G7 to G9 in every subgroup, but the widening is steepest for immigrant students in Gymnasium.

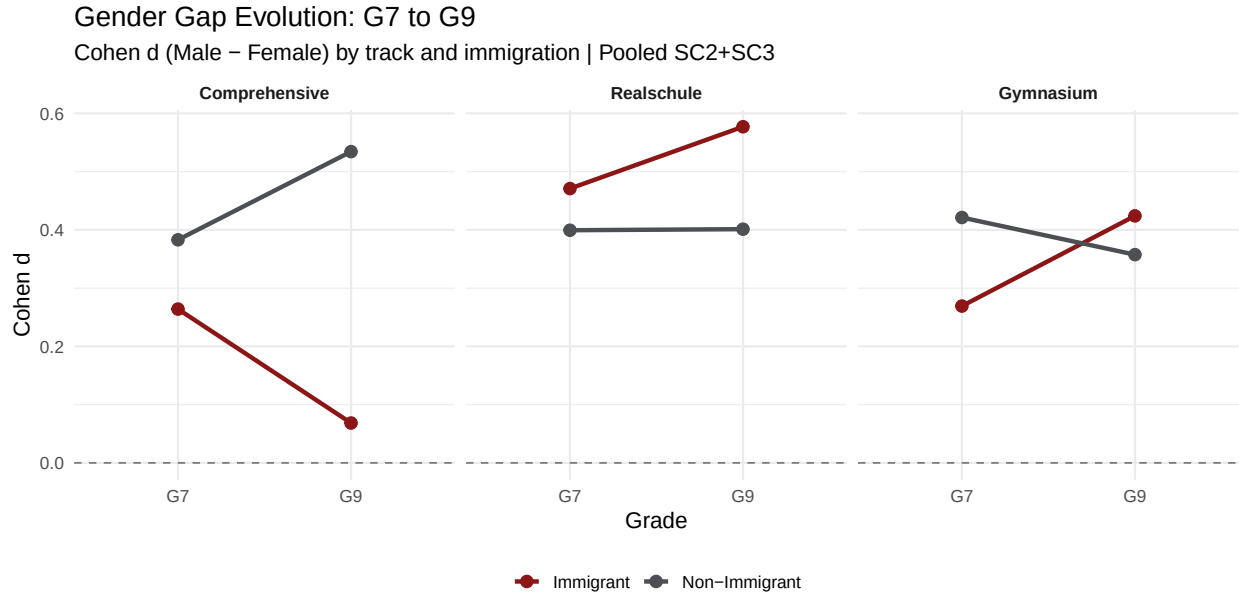


Figure 1: Gender gap evolution from G7 to G9 by track and immigration status. Cohen's d , defined as Male – Female, where positive values indicate male advantage. The gap widens in every track, but immigrant students in Gymnasium show the steepest divergence. Pooled SC2 + SC3, $N = 3,504$.

Tables 3–5 report the value-added regressions.

Table 3: Comprehensive Schools ($N = 424$)

	(1)	(2)	(3)	(4)
Math G7	0.568*** (0.034)	0.525*** (0.035)	0.523*** (0.035)	0.522*** (0.035)
Female	-0.211* (0.084)	-0.223** (0.082)	-0.315*** (0.093)	-0.437** (0.151)
Immigrant		-0.181 (0.100)	-0.401** (0.145)	-0.432** (0.148)
High SES		0.267** (0.090)	0.265** (0.089)	0.170 (0.130)
Female $\times$ Immigrant			0.402* (0.192)	0.454* (0.199)
Female $\times$ High SES				0.176 (0.172)
R^2	0.427	0.447	0.453	0.454
Adj. R^2	0.424	0.442	0.446	0.446
N	424	424	424	424

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Dependent variable: Math WLE at G9. Both cohorts pooled.

7.1 The gender gap widens across all tracks

The Female coefficient is negative and significant in Column 1 of every table: -0.211 in Comprehensive, -0.177 in Realschule, and -0.158 in Gymnasium, all at conventional significance levels. Girls who scored identically to boys at G7 end up between 0.16 and 0.21 standard deviations lower at G9. The coefficient is stable from Columns 1 to 2; adding immigration status and SES barely moves it, confirming that the gap is not a compositional artifact. It develops within demographic groups.

7.2 Immigration moderates the gap, but differently by track

The key finding is in Column 3. In Gymnasium, as reported in Table 5, the Female $\times$ Immigrant interaction is -0.184 and statistically significant. Immigrant boys in Gymnasium actually outperform non-immigrant boys, with a positive and highly significant Immigrant

Table 4: Realschule ($N = 780$)

	(1)	(2)	(3)	(4)
Math G7	0.557*** (0.027)	0.543*** (0.027)	0.543*** (0.027)	0.543*** (0.027)
Female	-0.177** (0.055)	-0.177** (0.054)	-0.157** (0.060)	-0.190* (0.089)
Immigrant		-0.113 (0.072)	-0.057 (0.101)	-0.062 (0.101)
High SES		0.128* (0.054)	0.129* (0.054)	0.104 (0.073)
Female $\times$ Immigrant			-0.112 (0.140)	-0.100 (0.142)
Female $\times$ High SES				0.054 (0.109)
R^2	0.388	0.395	0.395	0.396
Adj. R^2	0.386	0.392	0.392	0.391
N	780	780	780	780

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Dependent variable: Math WLE at G9.
Both cohorts pooled.

main effect of +0.223, consistent with positive selection into the academic track. But immigrant girls do not share this advantage: the interaction wipes out the immigrant premium for girls and adds to the gender gap. The total Female effect for immigrant girls in Gymnasium is $-0.121 + (-0.184) = -0.305$, roughly double the gap for non-immigrant girls. This is the “caught between cultures” dynamic: the high-aspiration academic environment amplifies identity conflict for immigrant girls.

In Comprehensive schools, as reported in Table 3, the pattern reverses. The Female $\times$ Immigrant interaction is +0.402 and statistically significant. Immigrant girls are relatively protected from the gender gap that non-immigrant girls experience. The total Female effect for immigrant girls is $-0.315 + 0.402 = +0.087$, effectively zero, or even slightly positive. The lower-pressure Comprehensive environment does not generate the same identity conflict. The Immigrant main effect is -0.401 , so immigrants generally perform worse in Comprehensive, but the gender gap within immigrants is near zero.

Table 5: Gymnasium ($N = 2,300$)

	(1)	(2)	(3)	(4)
Math G7	0.572*** (0.016)	0.571*** (0.017)	0.572*** (0.017)	0.572*** (0.017)
Female	-0.158*** (0.037)	-0.159*** (0.037)	-0.121** (0.042)	-0.183* (0.086)
Immigrant		0.129** (0.046)	0.223*** (0.065)	0.218*** (0.066)
High SES		0.209*** (0.046)	0.206*** (0.046)	0.168** (0.065)
Female $\times$ Immigrant			-0.184* (0.091)	-0.173 (0.091)
Female $\times$ High SES				0.075 (0.091)
R^2	0.369	0.376	0.377	0.378
Adj. R^2	0.369	0.375	0.376	0.376
N	2,300	2,300	2,300	2,300

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Dependent variable: Math WLE at G9. Both cohorts pooled.

In Realschule, as reported in Table 4, the interaction is small and insignificant at -0.112 . The intermediate track produces a uniform gender gap regardless of immigration status.

7.3 SES does not independently moderate the gap

The Female $\times$ High SES interaction is positive but insignificant in all three tracks, ranging from 0.054 in Realschule to 0.176 in Comprehensive. Household cultural capital, as proxied by books, does not significantly buffer girls against the widening gap once immigration is controlled. The action is in immigration status, not SES.

7.4 Occupational aspirations: immigrant premium is real

If the glass ceiling operates through aspirations, it should show up not only in test scores but in what students want to become. I regress realistic occupational aspiration, measured by

the SIOPS-88 prestige scale ranging from 13 to 78, on G7 math, gender, immigration, SES, and the Female $\times$ Immigrant interaction, separately by track. Table 6 reports the results.

Table 6: Occupational Aspiration by Track, SIOPS Prestige

	Comprehensive			Realschule			Gymnasium		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Math G7	2.42*** (0.73)	2.91*** (0.75)	2.86*** (0.75)	0.32 (0.54)	0.68 (0.54)	0.66 (0.55)	1.22*** (0.30)	1.39*** (0.30)	1.39*** (0.30)
Female	4.12* (1.83)	4.14* (1.79)	3.10 (2.02)	4.69*** (1.07)	4.63*** (1.05)	4.36*** (1.15)	2.43*** (0.69)	2.53*** (0.68)	2.34** (0.77)
Immigrant		8.49*** (2.18)	5.66 (3.38)		5.60*** (1.44)	4.76* (2.06)		4.19*** (0.83)	3.72** (1.19)
High SES		0.62 (1.96)	0.67 (1.96)		0.64 (1.06)	0.65 (1.06)		2.37** (0.81)	2.38** (0.81)
Female $\times$ Immig			4.69 (4.26)			1.62 (2.82)			0.90 (1.63)
R^2	0.051	0.105	0.109	0.036	0.064	0.065	0.017	0.037	0.037
N	264	264	264	517	517	517	1,446	1,446	1,446

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. DV: SIOPS-88 occupational prestige, scale 13–78. SC2 wave 9, SC3 wave 5. Pooled.

Two patterns are noteworthy. First, the immigrant aspiration premium is real: the Immigrant main effect is positive and highly significant in all three tracks in Column 2, ranging from +4.19 in Gymnasium to +8.49 in Comprehensive. Immigrant students aim for more prestigious occupations than natives, conditional on math ability. This confirms the immigrant aspiration paradox documented by Feliciano and Lanuza (2017): immigrant families transmit higher ambitions even when their children face structural disadvantages.

Second, the Female $\times$ Immigrant interaction does not reach significance in any track, though the coefficient magnitude is informative. In Comprehensive the interaction is +4.69, rather positive though highly variant. Immigrant girls, on average, aim higher than immigrant boys. In Gymnasium it is near zero at +0.90: the immigrant aspiration premium is equally shared between genders. The diminished motivation among girls in the gymnasium is reflected in their academic skill development.

8 Matched sample and quasi-experimental evidence

The value-added regressions condition on G7 scores, but students are already sorted by G7. A sharper test uses pre-tracking scores. Using the SC2 cohort, I match students on G4 math scores across tracks within the Q10–Q90 common support, trimming the Gymnasium upper tail. The matched sample contains 626 students with balanced G4 means: $F(2, 623) = 0.151$, $p = 0.860$.

Figure 2 shows the divergence: students who scored identically at G4 end up 0.57 SD apart by G9, depending on track assignment. This seems like a *prima facie* treatment effect of tracking on math growth. What it does show is that the comprehensive average is on a par with the real school average. Parents who opt out of the tracking system are usually recommended for a real school instead of a gymnasium. Therefore, it is reasonable to conjecture that parents perceive different tracks as having different effects on student performances.

Figure 3 reports the Female and Female $\times$ Immigrant coefficients from the value-added regression on the matched sample. The Gymnasium Female coefficient doubles to -0.355 , highly significant, relative to the pooled estimate, confirming that the pooled sample understates the gap. The Female $\times$ Immigrant interaction in matched Gymnasium is -0.280 , consistent with the glass ceiling hypothesis. Moreover, the comprehensive coefficient is positive, albeit with noise and lack of power. Tables 7–9 provide the full regression output.

9 Structural Estimation: Penn’s Threshold in Comprehensive vs. Gymnasium

Penn (2017) predicts that a class-specific aspiration threshold $\bar{s}$ divides students into two regimes: above it, interpersonal comparison motivates effort and scores converge with low persistence γ_{above} ; below it, the status gap discourages investment and scores lock in with

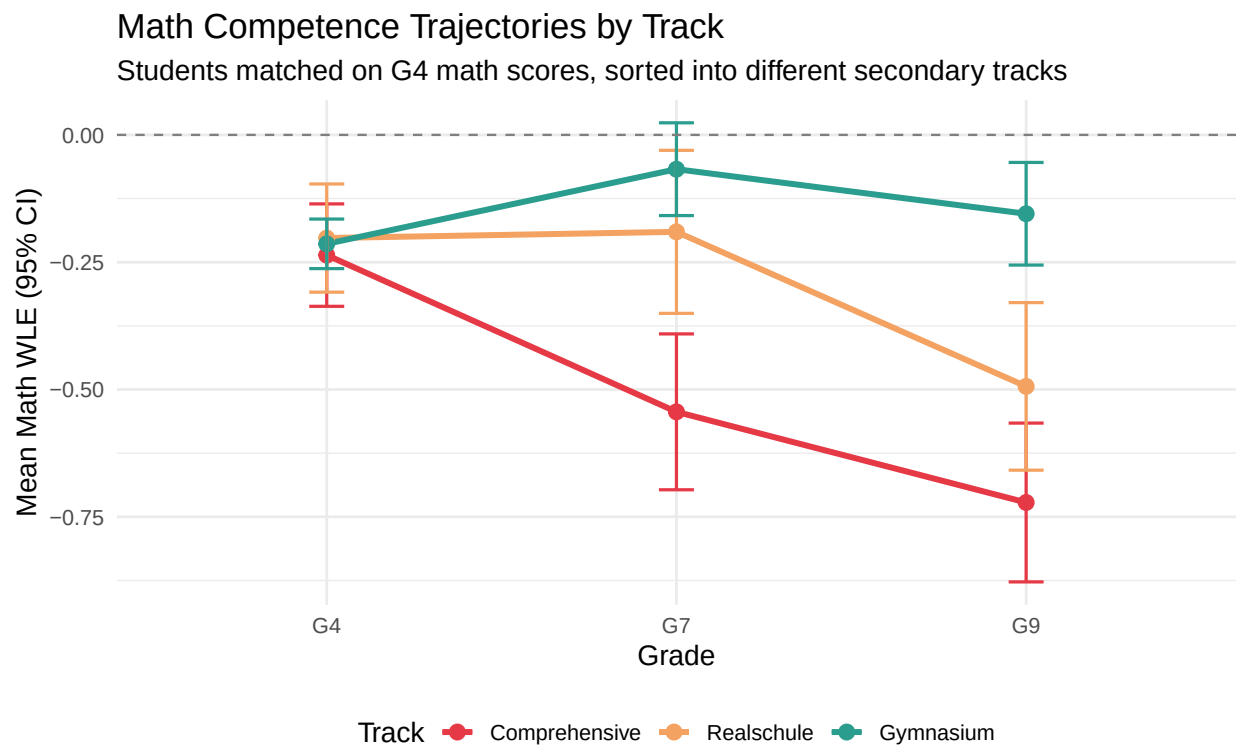


Figure 2: Math competence trajectories by track in the matched sample. Students are balanced on G4 math scores by construction. Gymnasium students maintain or improve; Comprehensive students lose ≈ 0.5 SD by G9. SC2 only, $N = 626$.

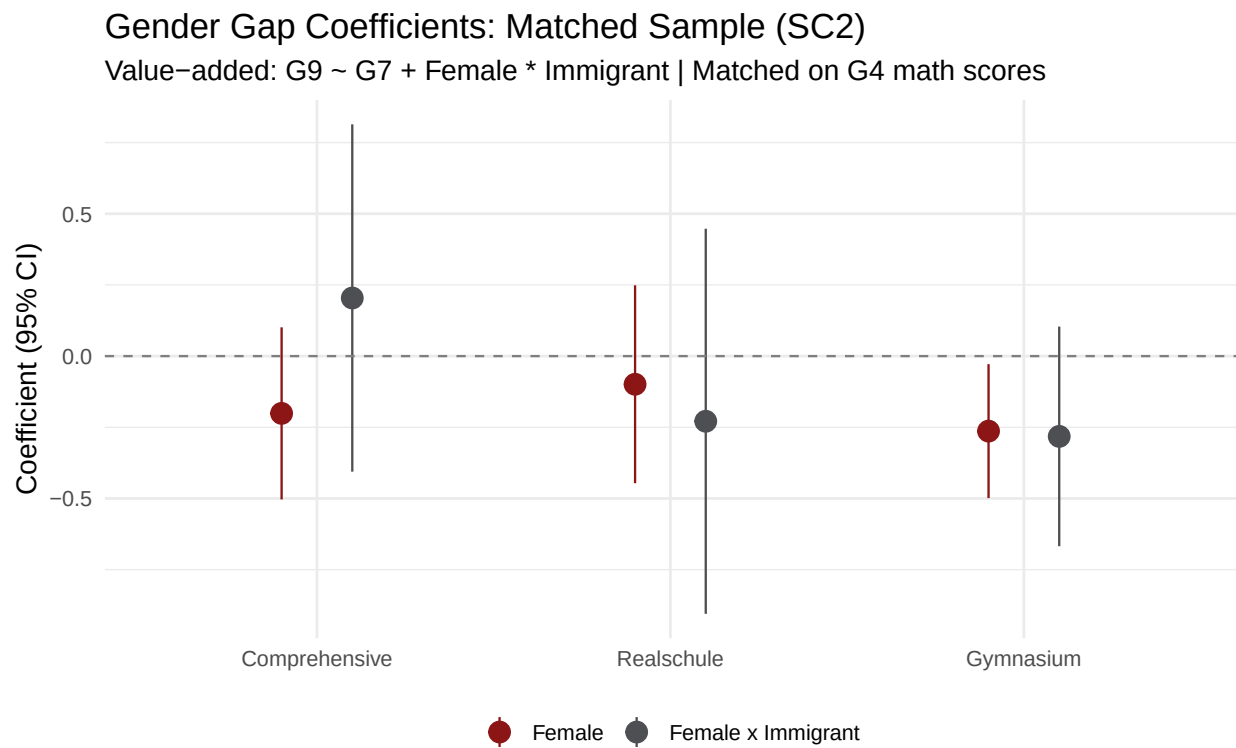


Figure 3: Value-added coefficients on Female and Female $\times$ Immigrant in the matched sample, SC2. Reference: male non-immigrant. The Gymnasium Female coefficient of -0.36 is more than double the pooled estimate. Error bars: 95% CI.

Table 7: Comprehensive Schools, Matched Sample ($N = 180$)

	(1)	(2)	(3)	(4)
Math G7	0.577*** (0.064)	0.558*** (0.064)	0.553*** (0.064)	0.554*** (0.065)
Female	-0.154 (0.133)	-0.143 (0.132)	-0.190 (0.152)	-0.149 (0.235)
Immigrant		-0.183 (0.154)	-0.287 (0.227)	-0.279 (0.230)
High SES		0.245 (0.134)	0.243 (0.134)	0.276 (0.197)
Female $\times$ Immigrant			0.191 (0.307)	0.180 (0.312)
Female $\times$ High SES				-0.061 (0.269)
R^2	0.346	0.366	0.367	0.367
N	180	180	180	180

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Matched on G4 math (SC2 only). DV: Math WLE G9.

Table 8: Realschule, Matched Sample ($N = 130$)

	(1)	(2)	(3)	(4)
Math G7	0.476*** (0.081)	0.473*** (0.080)	0.468*** (0.080)	0.468*** (0.080)
Female	-0.166 (0.150)	-0.141 (0.150)	-0.083 (0.173)	-0.262 (0.275)
Immigrant		-0.063 (0.168)	0.049 (0.239)	0.038 (0.239)
High SES		0.323* (0.156)	0.323* (0.156)	0.198 (0.216)
Female $\times$ Immigrant			-0.224 (0.337)	-0.199 (0.339)
Female $\times$ High SES				0.263 (0.313)
R^2	0.233	0.261	0.263	0.267
N	130	130	130	130

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Matched on G4 math (SC2 only). DV: Math WLE G9.

Table 9: Gymnasium, Matched Sample ($N = 316$)

	(1)	(2)	(3)	(4)
Math G7	0.404*** (0.057)	0.394*** (0.058)	0.397*** (0.058)	0.399*** (0.058)
Female	-0.355*** (0.095)	-0.361*** (0.096)	-0.259* (0.120)	-0.330 (0.234)
Immigrant		-0.077 (0.098)	0.069 (0.142)	0.065 (0.142)
High SES		0.137 (0.119)	0.135 (0.118)	0.082 (0.191)
Female $\times$ Immigrant			-0.280 (0.196)	-0.274 (0.197)
Female $\times$ High SES				0.085 (0.241)
R^2	0.195	0.200	0.205	0.206
N	316	316	316	316

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Matched on G4 math (SC2 only). DV: Math WLE G9.

high persistence γ_{below} . The reduced form is a piecewise linear model:

$$s_{t+1}^i = \begin{cases} a + \gamma_1 \cdot s_t^i + \epsilon_i & \text{if } s_t^i > \bar{s} \\ a + \gamma_2 \cdot s_t^i + \epsilon_i & \text{if } s_t^i \leq \bar{s} \end{cases} \quad (2)$$

with Penn's prediction $\gamma_1 < \gamma_2$. I estimate $\bar{s}$ by Hansen (2000) grid search on the full unmatched pooled sample for the G7→G9 transition, separately by track, gender, and immigration status. This produces a 2×4 typology: two tracks, Comprehensive and Gymnasium, crossed with four demographic cells of Native Male, Immigrant Male, Native Female, and Immigrant Female. Figure 4 reports all eight cells.

The eight panels tell a clean story. In seven of them, persistence is higher above the threshold than below, or the two regimes are roughly symmetric. Students who clear the aspiration cutoff see their effort compound; those below it are mobile, not trapped. This holds for native boys and native girls in both tracks, for immigrant boys in both tracks, and

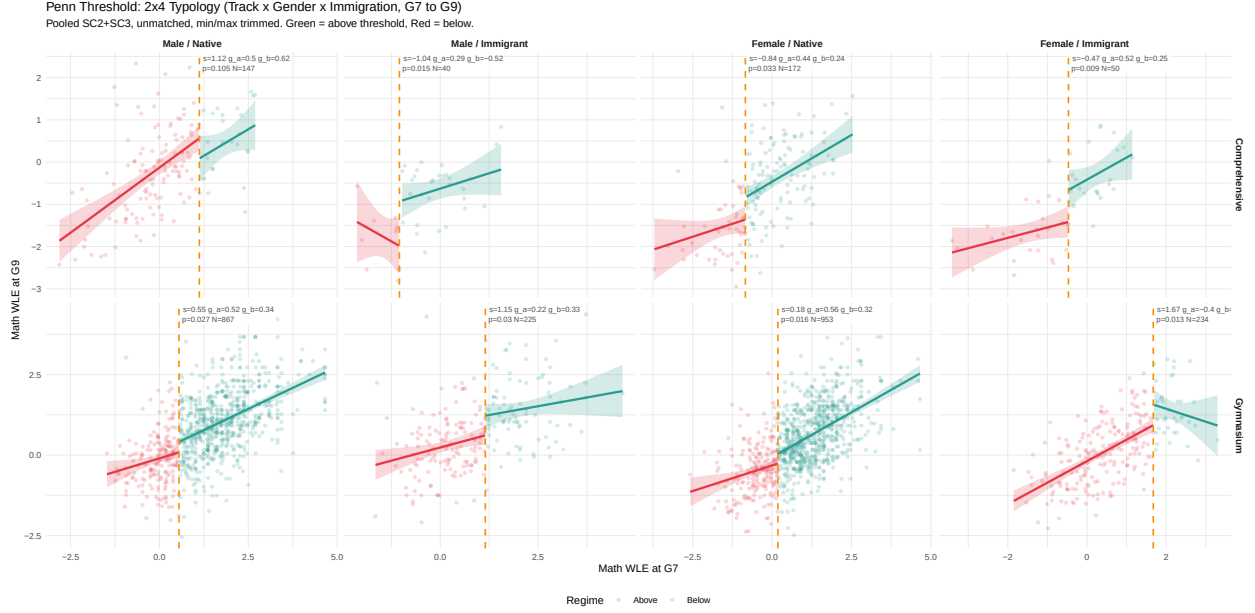


Figure 4: Penn threshold estimation: 2×4 typology of Track $\times$ Gender $\times$ Immigration, G7 $\rightarrow$ G9. Pooled SC2+SC3, unmatched, min/max trimmed. Each panel shows the piecewise regression with the estimated threshold in orange dashed. Green = above threshold; red = below. The aspiration trap, Penn's lock-in where the red slope is steeper than the green, appears in one cell only: Gymnasium Immigrant Female.

for immigrant girls in comprehensive. The threshold marks where effort starts to pay off.

One cell breaks the pattern. Gymnasium Immigrant Girls, with $N = 234$, are the only group that exhibits Penn's aspiration trap, and they exhibit it in its sharpest form: $\gamma_{\text{above}} = -0.402$, $\gamma_{\text{below}} = 0.668$, $F = 4.46$, $p = 0.013$. Below the threshold at $\bar{s} = 1.67$, scores lock in hard; a girl who falls behind at G7 stays behind at G9. Above the threshold, persistence is not just low but negative. The few immigrant girls who clear it actually regress. Effort above the cutoff does not compound, it dissipates. This is the structural signature of the glass ceiling. It is not a uniform drag on performance. It is a regime switch in which the normal reward to effort, the pattern that operates for everyone else, is reversed for immigrant girls in the academic track.

The contrast with Comprehensive Immigrant Girls is instructive despite a small $N = 50$. Their piecewise structure is significant at $F = 5.18$, $p = 0.009$, and goes the other way: $\gamma_{\text{above}} = 0.524$, $\gamma_{\text{below}} = 0.246$. Effort compounds above the threshold, no lock-in below.

Immigrant girls in the lower-pressure track see the same dynamics as everyone else.

Gymnasium Native Girls, with $N = 953$, confirm that this is not a gender effect on its own. Their pattern is $\gamma_{\text{above}} = 0.559$, $\gamma_{\text{below}} = 0.318$ at $p = 0.016$; effort compounds above, mobility below. This is indistinguishable from native boys. The aspiration trap requires the intersection of immigrant status and the academic track. Neither alone is sufficient.

The structural estimation localizes the glass ceiling with precision. It is not girls in general who face the trap. It is not immigrants in general. It is not the Gymnasium environment in general. It is the specific combination of traditional gender norms carried by immigrant families and high-pressure academic demands imposed by the Gymnasium environment that produces a regime switch in which effort above the threshold is punished rather than rewarded, and position below it is frozen rather than fluid. The glass ceiling is a piecewise discontinuity in the persistence function, operating at the intersection of gender, immigration, and institutional context.

10 Teacher-bias evidence at G7

To test whether teacher grading itself contributes to the gap, I regress the G7 Jahreszeugnis Note on the corresponding G7 IRT-scaled competence test, the immigrant indicator, the female indicator, and their interaction, separately by track and by subject. The Note is on the original German scale (1 best, 6 worst), so a positive coefficient on a group dummy means a worse Note conditional on objectively measured ability. The math Note is benchmarked against the G7 math IRT score (`mag7_sc1`); the German Note is benchmarked against the G7 reading IRT score (`reg7_sc1`). NEPS SC2 does not test reading at G9, so the bias test is run at G7 only, where both competence measures are directly observed.

Table ?? reports the four regressions. Two patterns dominate. First, the immigrant coefficient is positive in every column. It is significant in Gymnasium math at $+0.206$ ($p = 0.005$) and in Comprehensive German at $+0.406$ ($p = 0.005$), and positive but underpowered

in the other two. Conditional on the test score, immigrants receive worse Noten than natives across both tracks and both subjects. Second, the female coefficient is negative across the board, sharply so in Gymnasium German at -0.342 ($p < 0.001$). Girls collect a Note premium relative to test-scoring boys in every cell; only Gymnasium German is statistically distinguishable from zero. The Immigrant $\times$ Female interaction is small relative to the main effects, so immigrant girls inherit the additive sum of the two: an immigrant penalty plus a pro-girl premium. The interaction in Comprehensive German trends negative at -0.415 ($p = 0.058$), suggesting immigrant girls in that track are partially shielded from the immigrant penalty their male peers face.

The teachers push immigrant students down on the Note conditional on test score and pull girls up. Both directions hold across tracks. The two biases run in opposite directions for immigrant girls and partly cancel. This rules out teacher hostility toward girls as the source of the math gap the rest of the paper documents. If anything, teachers do the opposite. Where, then, does the academic underinvestment in immigrant girls come from? Not from a teacher who grades them down. The bias regressions point at the family side. The teacher's pro-girl Note premium rewards the social-compliance behavior that the family is already cultivating in the daughter. The premium is consistent with that choice, not the cause of any cost. What is missing is the academic insistence that would push the girl to over-invest in math the way her brother does. A lenient grade without an accompanying push leaves the academic dimension under-invested, and the girl falls behind on the IRT-scaled measure even as her Note holds up.

11 Discussion

The results line up with the glass ceiling hypothesis. Immigrant families carry an aspiration premium that benefits their children, but the premium is gendered, and the gendered differentiation activates only when the institutional environment demands sustained academic

	Math (1)	German (2)	Math (3)	German (4)
IRT	-0.368*** (0.023)	-0.223*** (0.019)	-0.234*** (0.041)	-0.176*** (0.035)
Immigrant	0.206** (0.073)	0.087 (0.064)	0.181 (0.164)	0.406** (0.145)
Female	-0.034 (0.059)	-0.342*** (0.051)	-0.080 (0.110)	-0.152 (0.098)
Immigrant $\times$ Female	-0.122 (0.102)	0.048 (0.090)	-0.287 (0.236)	-0.415 (0.218)
Constant	2.590*** (0.046)	2.625*** (0.038)	2.386*** (0.077)	2.451*** (0.069)
Track	Gymnasium	Gymnasium	Comprehensive	Comprehensive
IRT benchmark	Math (mag7)	Reading (reg7)	Math (mag7)	Reading (reg7)
Observations	1,304	1,257	313	316
R ²	0.175	0.156	0.106	0.129
Adjusted R ²	0.172	0.153	0.094	0.117
Residual Std. Error	0.860 (df = 1299)	0.743 (df = 1252)	0.855 (df = 308)	0.769 (df = 311)
F Statistic	68.803*** (df = 4; 1299)	57.789*** (df = 4; 1252)	9.118*** (df = 4; 308)	11.466*** (df = 4; 311)

Note: *p<0.05; **p<0.01; ***p<0.001

Note on original 1 (best) to 6 (worst) scale. Positive coefficient = penalty; negative = premium.

Math columns benchmark against G7 math IRT; German columns against G7 reading IRT.

Track assigned from school code at wave 9 (G7), back-filled with wave 7 / wave 10 if missing.

commitment. In Gymnasium that demand triggers the ceiling. The family’s traditional gender norms redirect the daughter’s investment toward gender-appropriate skills and reserve sustained academic insistence for the son. In comprehensive schools the demand is lower, the ceiling does not bind, and immigrant girls benefit from the same cultural emphasis on education that immigrant boys receive.

The occupational aspiration result anchors the aspirational reading. The glass ceiling does not show up only in math scores. It shows up in SIOPS occupational prestige. If the math gap were driven solely by merit and curriculum difficulty it would not extend to self-reported aspirations. It does, which means the mechanism runs through the student’s own expectation formation. This is the value-distortion channel that Penn’s model predicts.

The pattern maps onto Penn’s threshold model. In Penn’s framework a class-specific aspiration threshold s^* determines whether interpersonal comparison motivates or demobilizes. The glass ceiling is a gendered version of that threshold. For immigrant boys, the threshold is set by academic peers and is reachable through effort. For immigrant girls, the threshold is set by academic peers and competed against by the family’s marriage-market threshold, so the residual academic effort falls short. The girls who fall below the academic threshold enter Penn’s lock-in regime, not because they lack ability, but because the academic over-investment that boys can give freely is, for them, allocated elsewhere.

The null on SES reinforces the cultural reading. If the mechanism were material, with high-SES families simply investing more in daughters, then Female $\times$ High SES should be positive and significant. It is not. The glass ceiling is not about money or books in the household. It is about the cultural expectations that immigrant families bring and how those expectations interact with the institutional pressure of the academic track.

A caveat. The current analysis cannot causally distinguish family-side mechanisms, such as differential parental investment, from community-side mechanisms, such as peer pressure within immigrant social networks. The two are observationally consistent. The complementary evidence in the next sections, the matched sample and the structural estimation,

partially substitutes for the lack of an identification strategy.

12 Conclusion

This paper documents a glass ceiling in the math achievement of immigrant girls in the German tracking system and identifies the aspiration mechanism that produces it. The math gender gap widens across all three tracks during secondary school. Girls who started at the same level as boys end up meaningfully behind. The central result is that immigration status moderates the gap in opposite directions across tracks. In Gymnasium, immigrant girls fall behind faster than native girls; the high-aspiration academic environment widens the gap rather than narrowing it. In comprehensive schools the pattern reverses: immigrant girls are protected and the within-immigrant gender gap is roughly zero. The structural estimation of Penn's threshold model sharpens the picture. Out of eight demographic cells, Gymnasium immigrant girls are the only one exhibiting the aspiration trap, with effort above the threshold reversing rather than compounding and position below it freezing. Socioeconomic status does not moderate the gap in any track. The mechanism is cultural, not material.

The glass ceiling hypothesis pulls together four literatures. From aspiration economics (Ray, 2006; Genicot and Ray, 2017) it takes the idea that effort is non-monotonic in the aspiration gap. From identity economics (Akerlof and Kranton, 2002; Shayo, 2009) it takes the insight that group membership constrains behavioral prescriptions. From the immigrant aspiration paradox (Kao and Tienda, 1995; Feliciano and Lanuza, 2017; Salikutluk, 2016) it takes the regularity that immigrant families carry higher ambitions. From the tracking literature (Hanushek and Woessmann, 2006; Van de Werfhorst and Mijs, 2010) it takes the institutional mechanism that concentrates peer pressure and curricular demand. The synthesis is the gendered aspiration threshold. Immigrant families set high aspirations for all children, but the daughter's investment is allocated toward gender-appropriate skills, and the academic over-investment that boys give freely is, for her, allocated elsewhere. The

threshold is not biological, not material, and not imposed by the host society. It is carried within the immigrant family’s cultural expectations and triggered by institutional pressure.

The framework resolves a puzzle in the tracking literature. Dustmann et al. (2016) find zero average long-term effects of tracking, but the null masks compositional heterogeneity. Tracking simultaneously elevates immigrant boys, who benefit from the aspiration premium in a high-demand environment, and depresses immigrant girls, who hit the gendered ceiling in the same environment. The aggregate washes out, but the distributional consequences are large and policy-relevant. The result also extends the “caught between cultures” finding of Dahl et al. (2022) by giving it a structural form. It is not that opportunities generically backfire for immigrant girls. It is that the level of institutional demand determines whether the family’s cultural resources translate into academic over-investment or away from it.

The findings carry direct implications for education policy in tracked systems. Comprehensive schooling protects immigrant girls from the gendered aspiration trap. The implication is not that Gymnasium is harmful in itself, but that the interaction of Gymnasium-level demands with traditional gender norms produces the ceiling. Policies that reduce the salience of the identity conflict, such as mixed-ability classrooms within Gymnasium, mentoring programs aimed at immigrant girls, or explicit norm-setting that validates academic ambition for girls, may deactivate the ceiling without dismantling the track. The occupational aspiration results suggest that intervention at the aspiration level, not just the skill level, is required. Programs like the aspiration workshops in Orkin et al. (2023) and the female-leadership exposure of Beaman et al. (2012) may be more effective for immigrant girls than tutoring or curricular enrichment, because the binding constraint is not ability but cultural permission to aspire. The null on SES underscores that the glass ceiling is not a poverty story. Scholarships, free textbooks, and after-school programs will not on their own close the gap. The mechanism runs through cultural expectations, and policy has to engage the family’s understanding of gendered ambition if it is to work.

Several limitations warrant caution. The analysis cannot causally distinguish family-side

mechanisms, such as differential parental investment, from community-side mechanisms, such as peer pressure within immigrant social networks. Both fit the data. Disentangling them requires within-family variation through sibling fixed effects with opposite-sex pairs, which the current NEPS samples do not support with adequate power. “Immigrant” is also a coarse category. The NEPS migration indicator pools Turkish, Eastern European, Middle Eastern, and other origin groups that differ substantially in gender norms. The glass ceiling may be stronger for some origin groups than others, and a finer decomposition would sharpen the mechanism. The matched sample analysis relies on one cohort and the matched Gymnasium interaction is substantively large but does not reach conventional significance. The Penn threshold estimation in the Comprehensive Immigrant Female cell rests on a small N ; the point estimates are instructive but replication with larger samples is essential.

Three extensions follow. The mechanism can be tested directly by measuring parental aspirations for sons versus daughters within immigrant families across tracks, a prediction the hypothesis generates and the NEPS parental surveys can address. Quasi-experimental variation in track assignment, from state-level differences in the age of tracking, teacher recommendation policies, or randomized mentoring, can strengthen causal identification. The framework should be tested outside Germany. Any tracked system with a substantial immigrant population and origin-country variation in gender norms (Austria, the Netherlands, Switzerland) should exhibit the same pattern, because the mechanism depends on the interaction of institutional pressure and cultural gender expectations rather than on anything specific to Germany.

The glass ceiling is unshattered not because immigrant families lack ambition, but because the daughter’s share of that ambition is allocated toward gender-appropriate skills rather than toward academic over-investment. The aspiration premium that immigrant families carry is real, powerful, and beneficial. It is not gender-neutral, and the tracking system amplifies the asymmetry. Understanding the interaction is a prerequisite for designing institutions that serve all students equally.

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